

ERIC MANIRAGUHA

+250 788 746 696 ericmaniraguha@gmail.com

LinkedIn — GitHub — ORCID — Portfolio — Kigali, Rwanda

Data Engineering & ETL · SQL & Database Administration · Oracle PL/SQL · Python & PySpark · BI & Analytics · Data Quality & Governance · AI/ML · Cloud & DevOps

Professional Summary

Data Engineer and Database/Analytics professional with 10+ years of experience designing data systems, ETL pipelines, relational databases, BI solutions, and digital transformation initiatives across healthcare, humanitarian, research, and public-sector environments. MSc in Information Technology (Data Science & Machine Learning) from Carnegie Mellon University Africa. Strong expertise in SQL, Oracle PL/SQL, PostgreSQL, MySQL, Python, PySpark, Airflow, Power BI, data quality, database administration, and analytics. Experienced in translating complex data into reliable, decision-ready systems and insights.

Consulting & Advisory Experience

BrightSeed Hub Ltd.

November 2025 – Present

Database Engineering & Systems Analyst Consultant

Kigali, Rwanda

- Design, administer, and maintain scalable **database systems** supporting data-driven financial applications across 90+ units.
- Develop and optimize **SQL solutions, schemas, procedures, functions, indexes, ETL pipelines, and database performance**.
- Support **database security, user access, backup/recovery, monitoring, troubleshooting, and data protection** for business-critical systems.
- Lead **database migration, data transformation, cleansing, validation, and integration** for scalable business systems.

eFiche

July 2026 – October 2026

Digital Health Implementation Consultant

Kigali, Rwanda

- Led **data migration and database implementation** activities for EHR systems, including data validation, transformation, integrity checks, and SQL-based troubleshooting.
- Supported **frontend and backend localization** of the EHR platform, implementing and validating English–French translations across application interfaces, modules, validation messages, and system workflows.
- Worked with development and technical teams to validate **database structures, migrated data, application workflows, and API/backend responses**, ensuring consistency across the EHR platform.
- Supported **ETL, data quality, and interoperability** activities, helping ensure reliable exchange and accurate representation of healthcare data across system components.

Intouch Communications

May 2026 – August 2026

AI & Machine Learning Curriculum Design Consultant

Kigali, Rwanda

- Designed AI/ML curriculum covering supervised learning, deep learning, LLMs, NLP, RAG, MLOps, and applied AI systems.

Dussoft Engineering

May 2023 – November 2025

Data Engineering & Database Systems Consultant

Kigali, Rwanda

- Worked closely with **Database Administrators and Software Engineers** to design, maintain, and optimize relational databases supporting humanitarian and market intelligence systems.
- Applied **PostgreSQL, MySQL, SQL, Python, and Apache Airflow** for data integration, ETL pipelines, data quality, and reliable analytical data platforms.
- Supported **database performance optimization, indexing, schema design, monitoring, backup/recovery, and secure data access** across data-intensive applications.
- Implemented **Power BI and Kibana dashboards** for database-driven analytics, operational monitoring, supply-chain intelligence, and decision support.

Academic Experience

Adventist University of Central Africa – AUCA

September 2024 – Present

Assistant Lecturer – Oracle PL/SQL, Big Data Analytics & MIS

Kigali, Rwanda

- Teach Advanced Database Systems, Oracle PL/SQL, Big Data Analytics, and Management Information Systems; supervise projects in data science, BI, and data engineering.

University of Lay Adventists of Kigali – UNILAK

May 2026 – August 2026

Assistant Lecturer

Kigali, Rwanda

- Teach Database Programming, Oracle PL/SQL, Web Applications, E-Commerce, HCI, Digital Innovation, and related ICT courses with practical database development and systems training.

University of Kigali

January 2026 – February 2026

Assistant Lecturer

Kigali, Rwanda

- Delivered ICT Innovation, Intellectual Property, and emerging technology courses.

Technical Industry Experience

CyLab Africa / Upanzi Network

June 2023 – December 2025

Research Associate — Data Scientist & ICT Systems Analyst

Kigali, Rwanda

- Designed and administered **data platforms and pipelines** using Python, PySpark, PostgreSQL, Elasticsearch, and BI tools for Digital Public Infrastructure (DPI) and public-sector research.
- Performed **database modelling, SQL optimization, indexing, data validation, security, backup/recovery, performance monitoring, and data governance**.
- Applied **statistics, exploratory data analysis, performance metrics, predictive modelling, and machine learning** to generate evidence-based insights and evaluate system performance.
- Developed **Django, React, and REST API applications** incorporating AI/LLM, geospatial analytics, embeddings, and vector databases.

C4IR Rwanda

August 2023 – December 2023

AI Developer

Kigali, Rwanda

- Developed NLP/LLM document intelligence and semantic-search solutions using embeddings and vector databases for government information retrieval.

Dussoft Engineering

July 2019 – September 2022

Data Engineer

Kigali, Rwanda

- Designed and automated ETL pipelines using Python, Apache Airflow, SQL, and REST APIs for multi-source data integration, transformation, validation, and loading.
- Designed and optimized relational databases using SQL, stored procedures, views, triggers, indexing, window functions, normalization, and query tuning.
- Built cloud data warehousing and BI solutions using AWS, Amazon Redshift, Amazon S3, Power BI, and SQL.

Tech-Connect Business Park & IT Consultants

April 2015 – May 2019

ICT Systems & E-Business Solutions Consultant

Lusaka, Zambia

- Designed and supported enterprise ICT systems, relational databases, e-payment workflows, system integrations, and data quality controls across the software development lifecycle.
- Supported database administration activities including SQL development, database design, data validation, access controls, backup and recovery procedures, and troubleshooting of data-driven applications.
- Developed and integrated financial and e-business solutions, supporting reliable data processing, transaction workflows, system interoperability, and operational reporting.

Premier Bet Rwanda Limited

August 2012 – August 2015

Assistant Sales Manager & Retail Manager

Kigali, Rwanda

- Supported retail operations through reporting, performance tracking, operational dashboards, and data-driven decision-making.

Education

Carnegie Mellon University Africa

August 2021 – May 2024

MSc. Information Technology – Data Science & Machine Learning

Kigali, Rwanda

- GPA: 3.60 — Deep Learning, ML for Earth Observation, AI for Healthcare, Big Data Science, GIS, Applied ML.

Chreso University in Association with North End State University College

June 2013 – May 2017

BSc. Information Management

Lusaka, Zambia

- Grade: Merit — Database Management, SQL, Database Design, Statistics, Software Engineering, Information Systems, and Data Management.

Recent Projects

DRA VANUA Group Ltd

2026

ERP Application — Data Engineering & Systems Development

Kigali, Rwanda

- Led the **ERP project end-to-end**, from requirements gathering, system and database design to development, testing, deployment, and production support.
- Designed and developed a **MySQL ERP data platform** integrating sales, inventory, HR, payroll, finance, assets, bookings, and customer data.
- Developed **REST APIs, Sequelize models/migrations, RBAC, audit logging, and backup/recovery**.
- Built **ETL workflows, BI dashboards, and KPI reporting** for financial, HR, inventory, and operational analytics.
- Implemented **Grafana monitoring, activity-log analytics, data quality, indexing, and database security**.
- Deployed **Dockerized infrastructure** with automated database migrations and production-ready architecture.

Repository: [View Here](#)

HR Data Integration Pipeline — Data Engineering Project

2026

- Built a Python ETL pipeline integrating CSV, JSON, XLSX, and XML employee data with schema harmonization, salary normalization, department mapping, data quality validation, and duplicate detection.
- Produced an 18,200-record Golden Employee Dataset, identified 30 ghost employees, generated HR match-review reports, and created analytical visualizations with automated tests.

Repository: [View Here](#)

Humanitarian Market Intelligence Dashboard — Dussoft Engineering

2019 – 2022

- Designed and administered relational databases using database modelling, normalization, tables, views, stored procedures, functions, triggers, constraints, indexing, and SQL query optimization to support reliable market and supply-chain data management.
- Developed ETL and data integration workflows using Python, SQL, Apache Airflow, and AWS S3, integrating multiple data sources into analytical datasets and Power BI dashboards for food-price monitoring, predictive analytics, and humanitarian decision support.

Results: [View Here](#)

GeoScalable Malaria Surveillance Analytics — Kamabuye Health Center

2024 – 2025

- Developed geospatial data pipelines and GIS dashboards using MongoDB, Django, Airflow, PostgreSQL, Elasticsearch, and Kibana for malaria surveillance and hotspot analysis.

AI-Powered Document Navigator — C4IR Rwanda

2023

- Developed an NLP/LLM-powered document retrieval platform using document chunking, text preprocessing, embeddings, semantic search, and Retrieval-Augmented Generation (RAG) to enable intelligent government information discovery.
- Implemented a vector database for embedding storage and similarity search, with database indexing, metadata management, REST APIs, and Python-based processing pipelines to support efficient document retrieval and LLM-powered responses.

Certifications

- **Data Engineer Certification** – DataCamp [Certificate]
- **Big Data Fundamentals with PySpark** – DataCamp, Oct 2026 [Certificate]
- **Kubernetes and Cloud Native Associate (KCNA)** – Linux Foundation [Certificate]
- **Full-Stack Developer Certification** – Andela Rwanda [Certificate]
- **Additional Certifications (15+)** – AI/ML, Data Governance, Databases, Statistics, Data Science, Digital Transformation, FinTech, and Software Development. [Portfolio]

Technical Skills

- **Database Administration:** Oracle, PL/SQL, PostgreSQL, MySQL, SQL Server, MongoDB, Elasticsearch, Amazon Redshift; database configuration, schema management, data modelling, procedures, functions, triggers, indexing, query optimization, performance tuning, capacity planning, and troubleshooting.
- **Database Operations & Reliability:** Database monitoring, health checks, availability management, high availability, backup and recovery, disaster recovery (DR), business continuity, database upgrades, patching, migration, replication, incident troubleshooting, and operational documentation.
- **Database Security & Governance:** Role-based access control (RBAC), privileges and permissions, database auditing, data protection, data quality, data governance, metadata management, compliance controls, security best practices, and access management.
- **Monitoring & Observability:** Grafana, Kibana, database performance monitoring, system health monitoring, metrics, dashboards, alerting, log analysis, performance diagnostics, capacity monitoring, and proactive issue detection.
- **Data Engineering:** Python, PySpark, Pandas, ETL/ELT, Apache Airflow, Kafka, Data Lakes, Data Warehousing, Data Integration, Data Quality, dbt, and data pipeline development.
- **Analytics & AI:** SQL analytics, Power BI, Statistics, Exploratory Data Analysis (EDA), Predictive Analytics, Scikit-learn, TensorFlow, Deep Learning, and LLMs.
- **Software, Cloud & DevOps:** Django, Node.js, React, REST APIs, AWS, Docker, Kubernetes, CI/CD, Infrastructure Monitoring, and MLOps.
- **GIS & Digital Transformation:** ArcGIS Pro, Google Earth Engine, Spatial Analytics, Data Governance, Systems Integration, Interoperability, and Digital Transformation.

References

Available upon request.